PRODUCT IDENTITY

A



- **Product:** *Hapee Toothpaste* (Mouth Defense - Explosive Menthol Red), 55 g sachet
- **Compound Name**: Sodium Monofluorophosphate (Na_2PO_3F) [Active Ingredient]
- **Why it matters:** Active anti-cavity agent printed on the label as 1.14% w/w (1,500 ppm fluoride); fluoride ions strengthen tooth enamel and defend against acid attacks
- **Student:** Charmhel Galve Abonal, Chemistry for Engineers (BS Computer Engineering)

Hapee Toothpaste (Mouth Defense - Explosive Menthol Red, 55 g): from a printed label to moles, molecules and formula units
[MOLE CONCEPT | ACTIVITY 02]

ATOM COUNT CARD

BASE QUANTITATIVE DATA

B

Quantity	Value	Source	Precision
Net Sachet Mass	55 g	Product Label	2 s.f. (nominal fill)
Active Ingredient	1.14% w/w (Na_2PO_3F)	Product Label	3 s.f.
Chemical Formula	Na_2PO_3F	PubChem (n.d.)	Exact subscripts
Atomic masses Na, P, O, F	22.990,30.974,15.999,18.998	CIAAW (2024)	Not limiting
Avogadro constant, NA	6.02214076×1023 mol−1	BIPM (2019)	Exact

Assumptions (labeled): Calculations are based on the active ingredient Sodium Monofluorophosphate (Na_2PO_3F) at 1.14% mass concentration within the 55 g sachet.

MOLAR MASS BUILD-UP (NA_2PO_3F)

C

Element	Atoms	g/mol each	g/mol	% by mass
Na	2	22.99	45.98	33.094
P	1	30.974	30.974	22.285
O	3	15.999	47.997	34.536
F	1	18.998	18.998	13.671
Na_2PO_3F	**7**	—	**138.949**	**100**

Addition rule: each contribution has 3 decimal places, so M keeps 3 decimal places. Percentages sum to 100.000% (check).

THE MOLE MAP: ONE 55 G SACHET

D

55 g → 0.627 G NA_2PO_3F → 0.00451 MOL NA_2PO_3F → 0.00451 MOL NA_2PO_3F → 2.7×10^{21} FORMULA UNITS NA_2PO_3F

Always pass through moles; grams never convert straight to particles.
Name the entity at every step (mol of what?).

MOLE-CONVERSION CALCULATIONS

E

1. Molar mass of Na_2PO_3F
2(22.990) + 1(30.974) + 3(15.999) + 1(18.998)
= **138.949 g/mol**

2. Label bridge: percentage → mass of active compound in sachet
55 g paste × (1.14 g Na_2PO_3F / 100 g paste)
= 0.627 g
→ **6.3×10^{-1} g Na_2PO_3F**

3. Mass → moles
0.627 g × (1 mol / 138.949 g)
= 0.004512 mol
→ **4.5×10^{-3} mol Na_2PO_3F**

4. Moles → formula units
0.004512 mol × ($6.022\,140\,76 \times 10^{23}$ formula units / 1 mol)
= 2.717×10^{21} formula units
→ **2.7×10^{21} formula units Na_2PO_3F**

5. Atoms of one element inside the formula (Fluorine atoms)
2.717×10^{21} formula units × (1 F atom / 1 formula unit)
= 2.717×10^{21} F atoms
→ 2.7×10^{21} F atoms

6. Percent composition + two-route check (Fluorine)
% F = (18.998 g/mol ÷ 138.949 g/mol) × 100%
= 13.671%

Route A:
0.627 g × 0.13671 = 0.08572 g

Route B:
0.004512 mol × (1 mol F / 1 mol Na_2PO_3F) × (18.998 g / 1 mol F)
= 0.08572 g

Both routes:
8.6×10^{-2} g F ✓ agree

7. Reverse: from % composition to the formula
In 100 g:
Na: 33.094 ÷ 22.990 = 1.439 mol
P: 22.285 ÷ 30.974 = 0.719 mol
O: 34.536 ÷ 15.999 = 2.159 mol
F: 13.671 ÷ 18.998 = 0.719 mol

Divide by 0.719:
2.00 : 1.00 : 3.00 : 1.00

Therefore:
Na_2PO_3F *(empirical = molecular, n = 1.)*

8. Reverse: how much paste is one mole?
1.00 mol × (138.949 g / 1 mol) × (100 g paste / 1.14 g Na_2PO_3F)
= 12188.5 g paste
→ **1.2×10^{4} g paste**

SCALE-UP: FROM ONE SACHET TO A STORE INVENTORY

F

1. ONE SACHET	2. ONE CARTON, 120 SACHETS	3. REGIONAL SUPPLY, 12,000 SACHETS	4. REVERSE
55 g Hapee sachet	Assume 120 sachets per display box	Assume 12,000 sachets across stores	One mole of active Na_2PO_3F compound is inside
4.5×10^{-3} mol	0.542 mol	54.2 mol	2.2×10^{2} sachets
2.7×10^{21} formula units	120 × 0.00451 mol = 0.542 mol	12,000 × 0.00451 mol = 54.2 mol	of toothpaste: about 221 sachets

SIGNIFICANT-FIGURE CHECK

G

Calculator: 2.7173×10^{21} formula units
Reported: 2.7×10^{21} formula units

- Limits: 55 g sachet mass (2 s.f.). Exact: N_a, subscripts, sachet counts.
- Guard digits kept until the final line of every calculation.

ONE-NUMBER TAKEAWAY

"Even though a single 55 g sachet of Hapee Toothpaste contains only about 0.63 g of active sodium monofluorophosphate, it still delivers roughly 2.7×10^{21} formula units to defend your teeth against cavities."

H

REFERENCES (APA 7TH EDITION)

- Bureau International des Poids et Mesures. (2019). The International System of Units (SI) (9th ed.). https://www.bipm.org/en/publications/si-brochure
- Commission on Isotopic Abundances and Atomic Weights. (2024). Abridged standard atomic weights. International Union of Pure and Applied Chemistry. https://www.ciaaw.org/abridged-atomic-weights.htm
- National Center for Biotechnology Information. (n.d.). PubChem Compound Summary for CID 24413, Sodium monofluorophosphate. Retrieved October 17, 2026, from https://pubchem.ncbi.nlm.nih.gov/compound/Sodium-monofluorophosphate
- Lamoiyan Corporation. (n.d.). Hapee Toothpaste (Mouth Defense, Explosive Menthol Red), 55 g sachet [Product label]. Photograph by Charmhel Galve Abonal, October 17, 2026